

Problem

Use mesh analysis to find I_o in Fig. 10.28 (for Example 10.10).

Example 10.10:

Obtain current I_o in Fig. 10.28 using Norton's theorem.

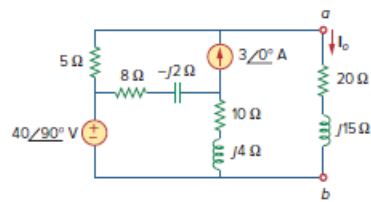


Figure 10.28
For Example 10.10.

Solution:

Our first objective is to find the Norton equivalent at terminals a - b . Z_N is found in the same way as Z_{Th} . We set the sources to zero as shown in Fig. 10.29(a). As evident from the figure, the $(8 - j2)$ and $(10 + j4)$ impedances are short-circuited, so that

$$Z_N = 5 \Omega$$

To get I_N , we short-circuit terminals a - b as in Fig. 10.29(b) and apply mesh analysis. Notice that meshes 2 and 3 form a supermesh because of the current source linking them. For mesh 1,

$$-j40 + (18 + j2)I_1 - (8 - j2)I_2 - (10 + j4)I_3 = 0 \quad (10.10.1)$$

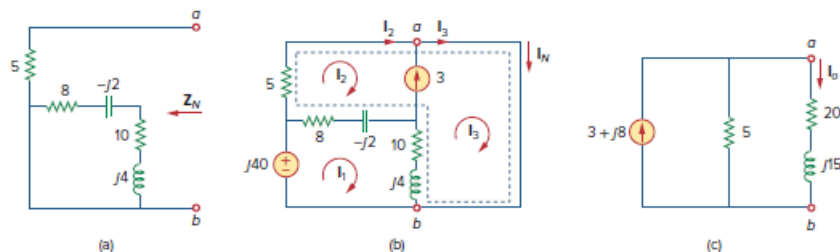


Figure 10.29
Solution of the circuit in Fig. 10.28: (a) finding Z_N , (b) finding I_N , (c) calculating I_o .

For the supermesh,

$$(13 - j2)I_2 + (10 + j4)I_3 - (18 + j2)I_1 = 0 \quad (10.10.2)$$

At node a , due to the current source between meshes 2 and 3,

$$I_3 = I_2 + 3 \quad (10.10.3)$$

Adding Eqs. (10.10.1) and (10.10.2) gives

$$-j40 + 5I_2 = 0 \quad \Rightarrow \quad I_2 = j8$$

From Eq. (10.10.3),

$$I_3 = I_2 + 3 = 3 + j8$$

The Norton current is

$$I_N = I_3 = (3 + j8) \text{ A}$$

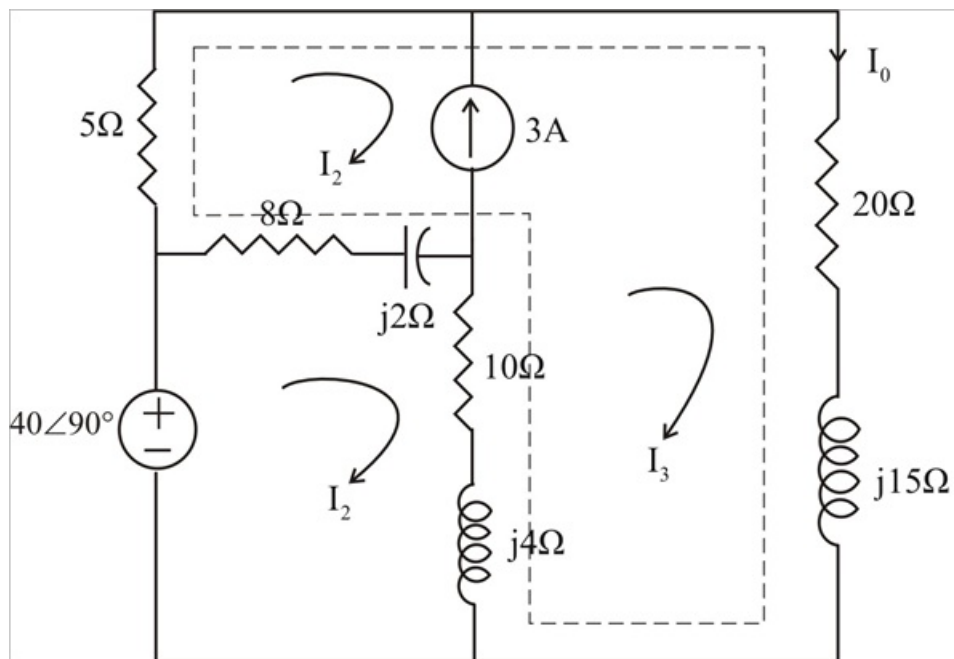
Figure 10.29(c) shows the Norton equivalent circuit along with the impedance at terminals a - b . By current division,

$$I_o = \frac{5}{5 + 20 + j15} I_N = \frac{3 + j8}{5 + j3} = 1.465 \angle 38.48^\circ \text{ A}$$

Step-by-step solution

Step 1 of 2

The circuit is shown below.



Step 2 of 2

For Mesh 1,

$$-j40 + (18+j2)I_1 - (8-j2)I_2 - (10+j4)I_3 = 0 \rightarrow (1)$$

For the Supermesh,

$$(13-j2)I_2 + (30+j19)I_3 - (18+j2)I_1 = 0 \rightarrow (2)$$

$$\text{Also, } I_2 = I_3 - 3 \rightarrow (3)$$

Adding (1) & (2) and incorporating (3)

$$-j40 + 5(I_3 - 3) + (20 + j15)I_3 = 0$$

$$I_3 = \frac{3+j8}{5+j3} = 1.465 \angle 38.48^\circ$$

$$I_0 = I_3 = 1.465 \angle 38.48^\circ \text{ A}$$